Robustness Appendix

Effects of Medicaid Expansion on Insurance Status, Emergency Department Payers, and Rural Hospital Finances

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This appendix accompanies the paper and reports the robustness checks summarized there. section A covers the state-level design, with the full set of specifications for every headline outcome, alternative estimators for staggered adoption, checks on the survey source and the comparison states, randomization inference, and additional event studies, including the Medicare placebo. section B covers the rural hospital design, with raw trends for every adoption cohort, pre-trend diagnostics, the within-state exposure design and its event studies, an alternative county-level exposure measure, and checks on the sample. Throughout, the preferred state-level specification is column (B) of the full tables, which includes state fixed effects, the full set of year fixed effects, and BEA region-by-year fixed effects.

A State-Level Design

A.1 Full Specification Tables

Table A.1 through Table A.6 report every specification for each headline outcome discussed in the paper, in the same format as the uninsured-rate table in the main text. Column (A) includes state fixed effects and year fixed effects; column (B), the preferred model, adds region-by-year effects; column (C) adds the poverty rate, the unemployment rate, and log median household income, which remain in every later column; column (D) adds an indicator for the two years before adoption; column (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (California, Connecticut, the District of Columbia, Minnesota, New Jersey, and Washington through early ACA expansions, and Delaware, Massachusetts, New York, and Vermont through earlier programs); and columns (F) and (G) add, in turn, the log Medicare enrollment rate and state-specific linear trends. Panel C of each table, where a count counterpart exists, reports a negative binomial model with state population as the exposure.

Three patterns hold across the headline outcomes. First, the two-way FE estimates in column (A) are close to the preferred estimates in column (B) for the coverage and ED outcomes, so adding region-by-year effects changes little. The exception is asthma inpatient stays, where column (A) is roughly 1.6 to 2.4 times column (B). Second, columns (B) through (F) are close to one another. The Medicaid enrollment estimate ranges from a 42 to a 44% increase unweighted, the Medicaid ED estimate from a 50 to a 58% increase, and the self-pay ED estimate from a 43 to a 49% decline. Third, state-specific linear trends in column (G) shrink every coverage and ED estimate somewhat (the asthma estimates grow) but leave all of the headline estimates significant.

One outcome needs more care. Asthma inpatient stays (Table A.5 and Table A.6) cover 2003–2017 and fewer state-years, because many states report small counts. They are hospital stays rather

than ED visits, because the HCUP state files report adult asthma by payer only for inpatient care. Medicaid-paid asthma stays rise by 19% unweighted, significant only at the 10 percent level, and by an insignificant 16% weighted, and uninsured asthma stays fall by 42 to 43%. The stacked event study in Table A.7 shows no divergence in uninsured asthma stays before adoption (-1.1% and 1.5% at three and two years before), but the estimates rest on noisy counts.

Table A.1: Log Medicaid Enrollment Rate — Difference-in-Differences Estimates

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Panel A: Unweighted OLS on log outcome							
Med. Exp.	0.3255***	0.3483***	0.3487***	0.3660***	0.3628***	0.3480***	0.2697***
	(0.0327)	(0.0410)	(0.0403)	(0.0461)	(0.0517)	(0.0406)	(0.0386)
1–2y pre				0.0609**	0.0469		
				(0.0279)	(0.0329)		
Panel B: Weighted OLS on log outcome (population)							
Med. Exp.	0.2359***	0.2800***	0.2745***	0.2872***	0.2842***	0.2744***	0.2206***
	(0.0515)	(0.0367)	(0.0364)	(0.0426)	(0.0455)	(0.0362)	(0.0299)
1–2y pre				0.0446	0.0261		
				(0.0307)	(0.0323)		
Panel C: Negative binomial on count outcome (population exposure)							
Med. Exp.	0.3162***	0.3299***	0.3273***	0.3424***	0.3294***	0.3271***	0.2335***
	(0.0302)	(0.0322)	(0.0317)	(0.0368)	(0.0398)	(0.0319)	(0.0258)
1–2y pre				0.0525**	0.0327		
				(0.0249)	(0.0279)		
State FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Region × year		Y	Y	Y	Y	Y	Y
Poverty, unemployment, log median income			Y	Y	Y	Y	Y
1–2y pre indicator				Y	Y		
Excludes early-coverage states					Y		
Log Medicare enr.						Y	
State linear trends							Y
Observations	969	969	969	969	779	969	969

NOTES: Each column reports the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) and, where included, the 1–2 year pre-treatment indicator from a separate state-year panel regression of Log Medicaid Enrollment Rate on the listed covariates. Column (A) includes state fixed effects and year fixed effects. (B) adds region × year interactions. (C) adds the state poverty rate, unemployment rate, and log median household income, which remain in (D) through (G). (D) further adds an indicator equal to one in the two years immediately preceding adoption. (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). (F) replaces the pre-adoption indicator with the log Medicare enrollment rate. (G) adds state-specific linear time trends in lieu of the controls in (D) through (F). All specifications cluster standard errors at the state level (in parentheses). Panel A reports unweighted OLS on the log outcome; Panel B weights by the state's average population over the sample period. Panel C reports a negative-binomial regression of the underlying count on the same covariates, using state population as the exposure variable; coefficients are log incidence-rate ratios and so are directly comparable to the OLS-on-log estimates. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.2: Log Private Health Insurance Enrollment Rate — Difference-in-Differences Estimates

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Panel A: Unweighted OLS on log outcome							
Med. Exp.	-0.0305***	-0.0269***	-0.0249***	-0.0244**	-0.0215*	-0.0254***	-0.0216***
	(0.0082)	(0.0094)	(0.0087)	(0.0104)	(0.0119)	(0.0083)	(0.0057)
1–2y pre				0.0017	0.0037		
				(0.0073)	(0.0086)		
Panel B: Weighted OLS on log outcome (population)							
Med. Exp.	-0.0269***	-0.0300***	-0.0248***	-0.0245***	-0.0230**	-0.0246***	-0.0240***
	(0.0097)	(0.0075)	(0.0069)	(0.0082)	(0.0093)	(0.0067)	(0.0057)
1–2y pre				0.0011	0.0029		
				(0.0061)	(0.0071)		
Panel C: Negative binomial on count outcome (population exposure)							
Med. Exp.	-0.0300***	-0.0301***	-0.0258***	-0.0258***	-0.0240***	-0.0256***	-0.0230***
	(0.0078)	(0.0070)	(0.0067)	(0.0079)	(0.0088)	(0.0064)	(0.0047)
1–2y pre				-0.0000	0.0016		
				(0.0056)	(0.0065)		
State FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Region × year		Y	Y	Y	Y	Y	Y
Poverty, unemployment, log median income			Y	Y	Y	Y	Y
1–2y pre indicator				Y	Y		
Excludes early-coverage states					Y		
Log Medicare enr.						Y	
State linear trends							Y
Observations	969	969	969	969	779	969	969

NOTES: Each column reports the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) and, where included, the 1–2 year pre-treatment indicator from a separate state-year panel regression of Log Private Health Insurance Enrollment Rate on the listed covariates. Column (A) includes state fixed effects and year fixed effects. (B) adds region × year interactions. (C) adds the state poverty rate, unemployment rate, and log median household income, which remain in (D) through (G). (D) further adds an indicator equal to one in the two years immediately preceding adoption. (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). (F) replaces the pre-adoption indicator with the log Medicare enrollment rate. (G) adds state-specific linear time trends in lieu of the controls in (D) through (F). All specifications cluster standard errors at the state level (in parentheses). Panel A reports unweighted OLS on the log outcome; Panel B weights by the state's average population over the sample period. Panel C reports a negative-binomial regression of the underlying count on the same covariates, using state population as the exposure variable; coefficients are log incidence-rate ratios and so are directly comparable to the OLS-on-log estimates. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.3: Log Medicaid ED Visits — Difference-in-Differences Estimates

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Panel A: Unweighted OLS on log outcome							
Med. Exp.	0.4192***	0.4557***	0.4255***	0.4270***	0.4230***	0.4035***	0.3659***
	(0.0796)	(0.0815)	(0.0739)	(0.0772)	(0.0829)	(0.0721)	(0.0773)
1–2y pre				0.0035	-0.0385		
				(0.0578)	(0.0629)		
Panel B: Weighted OLS on log outcome (population)							
Med. Exp.	0.4576***	0.4260***	0.3884***	0.4029***	0.3875***	0.3757***	0.3674***
	(0.0905)	(0.0942)	(0.0756)	(0.0808)	(0.0977)	(0.0799)	(0.1171)
1–2y pre				0.0411	-0.0357		
				(0.0805)	(0.0691)		
Panel C: Negative binomial on count outcome (population exposure)							
Med. Exp.	0.4136***	0.4517***	0.4225***	0.4244***	0.4204***	0.3993***	0.3588***
	(0.0755)	(0.0670)	(0.0610)	(0.0624)	(0.0624)	(0.0591)	(0.0591)
1–2y pre				0.0045	-0.0381		
				(0.0454)	(0.0464)		
State FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Region × year		Y	Y	Y	Y	Y	Y
Poverty, unemployment, log median income			Y	Y	Y	Y	Y
1–2y pre indicator				Y	Y		
Excludes early-coverage states					Y		
Log Medicare enr.						Y	
State linear trends							Y
Observations	403	403	403	403	319	403	403

NOTES: Each column reports the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) and, where included, the 1–2 year pre-treatment indicator from a separate state-year panel regression of Log Medicaid ED Visits on the listed covariates. Column (A) includes state fixed effects and year fixed effects. (B) adds region × year interactions. (C) adds the state poverty rate, unemployment rate, and log median household income, which remain in (D) through (G). (D) further adds an indicator equal to one in the two years immediately preceding adoption. (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). (F) replaces the pre-adoption indicator with the log Medicare enrollment rate. (G) adds state-specific linear time trends in lieu of the controls in (D) through (F). All specifications cluster standard errors at the state level (in parentheses). Panel A reports unweighted OLS on the log outcome; Panel B weights by the state's average population over the sample period. Panel C reports a negative-binomial regression of the underlying count on the same covariates, using state population as the exposure variable; coefficients are log incidence-rate ratios and so are directly comparable to the OLS-on-log estimates. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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Table A.4: Log Self-Pay/No-Charge ED Visits — Difference-in-Differences Estimates

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Panel A: Unweighted OLS on log outcome							
Med. Exp.	-0.4667***	-0.5737***	-0.5647***	-0.6060***	-0.6781***	-0.5589***	-0.3737**
	(0.0983)	(0.1989)	(0.1920)	(0.2104)	(0.2250)	(0.1995)	(0.1753)
1–2y pre				-0.0938	-0.0466		
				(0.0792)	(0.1017)		
Panel B: Weighted OLS on log outcome (population)							
Med. Exp.	-0.4771***	-0.5518**	-0.5390**	-0.5650**	-0.6656**	-0.5407**	-0.3997*
	(0.0754)	(0.2647)	(0.2518)	(0.2621)	(0.2993)	(0.2553)	(0.2216)
1–2y pre				-0.0739	0.0366		
				(0.1041)	(0.1153)		
Panel C: Negative binomial on count outcome (population exposure)							
Med. Exp.	-0.4755***	-0.5810***	-0.5728***	-0.6126***	-0.6895***	-0.5666***	-0.3805***
	(0.0891)	(0.1571)	(0.1515)	(0.1669)	(0.1672)	(0.1557)	(0.1291)
1–2y pre				-0.0906	-0.0545		
				(0.0627)	(0.0754)		
State FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Region × year		Y	Y	Y	Y	Y	Y
Poverty, unemployment, log median income			Y	Y	Y	Y	Y
1–2y pre indicator				Y	Y		
Excludes early-coverage states					Y		
Log Medicare enr.						Y	
State linear trends							Y
Observations	403	403	403	403	319	403	403

NOTES: Each column reports the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) and, where included, the 1–2 year pre-treatment indicator from a separate state-year panel regression of Log Self-Pay/No-Charge ED Visits on the listed covariates. Column (A) includes state fixed effects and year fixed effects. (B) adds region × year interactions. (C) adds the state poverty rate, unemployment rate, and log median household income, which remain in (D) through (G). (D) further adds an indicator equal to one in the two years immediately preceding adoption. (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). (F) replaces the pre-adoption indicator with the log Medicare enrollment rate. (G) adds state-specific linear time trends in lieu of the controls in (D) through (F). All specifications cluster standard errors at the state level (in parentheses). Panel A reports unweighted OLS on the log outcome; Panel B weights by the state's average population over the sample period. Panel C reports a negative-binomial regression of the underlying count on the same covariates, using state population as the exposure variable; coefficients are log incidence-rate ratios and so are directly comparable to the OLS-on-log estimates. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.5: Log Asthma Inpatient Stays, Medicaid Payers — Difference-in-Differences Estimates

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Panel A: Unweighted OLS on log outcome							
Med. Exp.	0.3700***	0.1703*	0.1569*	0.1100	0.1408	0.1492	0.2042**
	(0.0900)	(0.0937)	(0.0913)	(0.1003)	(0.1119)	(0.0903)	(0.0911)
1–2y pre				-0.1417*	-0.1307		
				(0.0711)	(0.0803)		
Panel B: Weighted OLS on log outcome (population)							
Med. Exp.	0.3490***	0.1442	0.1093	0.0467	0.0806	0.1109	0.1801**
	(0.0951)	(0.1009)	(0.0801)	(0.0865)	(0.1077)	(0.0840)	(0.0794)
1–2y pre				-0.1625**	-0.1638*		
				(0.0690)	(0.0816)		
Panel C: Negative binomial on count outcome (population exposure)							
Med. Exp.	0.3791***	0.1656**	0.1514**	0.1042	0.1302	0.1424*	0.1924***
	(0.0871)	(0.0750)	(0.0725)	(0.0795)	(0.0836)	(0.0714)	(0.0654)
1–2y pre				-0.1455**	-0.1352**		
				(0.0584)	(0.0629)		
State FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Region × year		Y	Y	Y	Y	Y	Y
Poverty, unemployment, log median income			Y	Y	Y	Y	Y
1–2y pre indicator				Y	Y		
Excludes early-coverage states					Y		
Log Medicare enr.						Y	
State linear trends							Y
Observations	472	472	472	472	386	472	472

NOTES: Each column reports the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) and, where included, the 1–2 year pre-treatment indicator from a separate state-year panel regression of Log Asthma Inpatient Stays, Medicaid Payers on the listed covariates. Column (A) includes state fixed effects and year fixed effects. (B) adds region × year interactions. (C) adds the state poverty rate, unemployment rate, and log median household income, which remain in (D) through (G). (D) further adds an indicator equal to one in the two years immediately preceding adoption. (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). (F) replaces the pre-adoption indicator with the log Medicare enrollment rate. (G) adds state-specific linear time trends in lieu of the controls in (D) through (F). All specifications cluster standard errors at the state level (in parentheses). Panel A reports unweighted OLS on the log outcome; Panel B weights by the state's average population over the sample period. Panel C reports a negative-binomial regression of the underlying count on the same covariates, using state population as the exposure variable; coefficients are log incidence-rate ratios and so are directly comparable to the OLS-on-log estimates. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.6: Log Asthma Inpatient Stays, Uninsured Payers — Difference-in-Differences Estimates

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Panel A: Unweighted OLS on log outcome							
Med. Exp.	-0.9405***	-0.5583**	-0.6143***	-0.5389**	-0.7002***	-0.6376***	-0.6682***
	(0.1656)	(0.2215)	(0.1556)	(0.2048)	(0.2263)	(0.1558)	(0.1542)
1–2y pre				0.1083	0.1139		
				(0.1276)	(0.1316)		
Panel B: Weighted OLS on log outcome (population)							
Med. Exp.	-1.1382***	-0.5447**	-0.5657***	-0.4428*	-0.6429**	-0.5808***	-0.5657***
	(0.2085)	(0.1999)	(0.1498)	(0.2388)	(0.2954)	(0.1493)	(0.1737)
1–2y pre				0.1539	0.1732		
				(0.1587)	(0.1696)		
Panel C: Negative binomial on count outcome (population exposure)							
Med. Exp.	-0.8984***	-0.5503***	-0.6053***	-0.5188***	-0.6883***	-0.6261***	-0.6661***
	(0.1527)	(0.1697)	(0.1201)	(0.1619)	(0.1765)	(0.1180)	(0.1193)
1–2y pre				0.1200	0.1246		
				(0.1005)	(0.1017)		
State FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Region × year		Y	Y	Y	Y	Y	Y
Poverty, unemployment, log median income			Y	Y	Y	Y	Y
1–2y pre indicator				Y	Y		
Excludes early-coverage states					Y		
Log Medicare enr.						Y	
State linear trends							Y
Observations	368	368	368	368	297	368	368

NOTES: Each column reports the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) and, where included, the 1–2 year pre-treatment indicator from a separate state-year panel regression of Log Asthma Inpatient Stays, Uninsured Payers on the listed covariates. Column (A) includes state fixed effects and year fixed effects. (B) adds region × year interactions. (C) adds the state poverty rate, unemployment rate, and log median household income, which remain in (D) through (G). (D) further adds an indicator equal to one in the two years immediately preceding adoption. (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). (F) replaces the pre-adoption indicator with the log Medicare enrollment rate. (G) adds state-specific linear time trends in lieu of the controls in (D) through (F). All specifications cluster standard errors at the state level (in parentheses). Panel A reports unweighted OLS on the log outcome; Panel B weights by the state's average population over the sample period. Panel C reports a negative-binomial regression of the underlying count on the same covariates, using state population as the exposure variable; coefficients are log incidence-rate ratios and so are directly comparable to the OLS-on-log estimates. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.2 Alternative Estimators for Staggered Adoption

When treatment is adopted at different times, a two-way fixed effects regression is a weighted average of many two-by-two comparisons, some of which use already-treated states as controls for later-treated ones; if effects grow over time, those comparisons can bias the estimate (Goodman-Bacon, 2021).

Table A.7 reports a stacked difference-in-differences model in the style of Wing et al. (2024). For each adoption cohort, I build a sub-experiment containing the cohort and only states that have not yet adopted, within three years before and two years after adoption, and then stack the sub-experiments. The two pre-adoption coefficients test for divergence before treatment. For the uninsured rate they are 3.4% and 3.7%, neither significant, and for Medicaid ED visits, self-pay ED visits, and both asthma series none is significant. A few small pre-adoption coefficients are significant at the 10 percent level or better, namely −2.0% for Medicaid enrollment three years before adoption and 1.0% for PHI two years before, each small next to the post-adoption estimates. After adoption, the effects grow over time, and the uninsured rate falls by 13% in the adoption year, 22% one year later, and 25% two years later.

Table A.8 compares the two-way fixed effects estimate with the Callaway and Sant'Anna (2021) average treatment effect on the treated, which compares each cohort only with never-adopting states and includes the poverty rate, unemployment rate, and log median household income as covariates; it does not include region-by-year effects. The two-way fixed effects column in this table is column (D) of the full tables. The Callaway and Sant'Anna (2021) estimates are somewhat smaller in magnitude than the preferred column (B) estimates for most outcomes, for example a 19% against a 29% decline for the uninsured rate and a 51% against a 58% rise for Medicaid ED visits, but every one has the same sign and is statistically significant. The gap does not come from leaving out region-by-year effects, since the two-way FE estimate without them, column (A), is a 28% decline. For the uninsured rate it comes in roughly equal parts from the earlier years of the sample, from the change of survey in 2013, both of which this estimator avoids by comparing each cohort with the year before adoption, and from its covariates.

Table A.7: Stacked DiD — Event-Study Coefficients

	Unins.	PHI	Mdcd Enr.	Mdcd ED	S-P ED	Asth. Mdcd	Asth. Unins.
$\tau = -3$	0.033	0.007	-0.020*	0.010	-0.032	0.045	-0.011
	(0.029)	(0.005)	(0.012)	(0.055)	(0.066)	(0.078)	(0.081)
$\tau = -2$	0.036	0.010*	-0.010	-0.014	-0.031	-0.026	0.015
	(0.024)	(0.005)	(0.006)	(0.032)	(0.057)	(0.065)	(0.057)
$\tau = 0$	-0.139***	-0.011***	0.175***	0.360***	-0.342***	0.278***	-0.568***
	(0.019)	(0.004)	(0.021)	(0.067)	(0.071)	(0.069)	(0.145)
$\tau = 1$	-0.243***	-0.022***	0.241***	0.465***	-0.482***	0.286***	-1.168***
	(0.033)	(0.006)	(0.028)	(0.082)	(0.102)	(0.104)	(0.173)
$\tau = 2$	-0.288***	-0.031***	0.267***	0.410***	-0.568***	0.448***	-1.065***
	(0.040)	(0.008)	(0.033)	(0.105)	(0.126)	(0.099)	(0.357)
Observations	528	528	528	338	338	312	251

NOTES: Each column reports event-study coefficients from a single Wing-Freedman-Hollingsworth stacked DiD regression of the indicated log outcome. The columns are *Unins.* = log uninsured rate; *PHI* = log private health insurance enrollment rate; *Mdcd Enr.* = log Medicaid enrollment rate; *Mdcd ED* = log adult Medicaid-paid ED visits per 100 people; *S-P ED* = log adult self-pay/no-charge ED visits per 100 people; *Asth. Mdcd* = log Medicaid-paid asthma inpatient stays per 100 people; *Asth. Unins.* = log uninsured asthma inpatient stays per 100 people. For each adoption cohort I build a sub-experiment of the cohort's states plus only-not-yet-treated controls within event time $[-3, +2]$ of treatment, then stack the sub-experiments and weight by the inverse treatment share within each stack so that each cohort contributes proportionally to its size. Reported coefficients are the interaction of the treated indicator with the indicated event-time indicator; $\tau = -1$ is the omitted reference. Each regression absorbs treatment-status and event-time fixed effects and clusters standard errors at the state level (in parentheses). Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.8: TWFE vs. Callaway-Sant'Anna ATT

	TWFE (D)	Callaway-Sant'Anna	Difference
Log Uninsured Rate	-0.3581	-0.2073	0.1507
	(0.0455)	(0.0578)	
Log PHI Enrollment Rate	-0.0244	-0.0300	-0.0056
	(0.0104)	(0.0106)	
Log Medicaid Enrollment Rate	0.3660	0.2230	-0.1430
	(0.0461)	(0.0347)	
Log Medicaid ED Visits	0.4270	0.4104	-0.0166
	(0.0772)	(0.1045)	
Log Self-Pay/No-Charge ED Visits	-0.6060	-0.3598	0.2462
	(0.2104)	(0.1683)	
Log Asthma Stays, Medicaid Payers	0.1100	0.2230	0.1131
	(0.1003)	(0.0720)	
Log Asthma Stays, Uninsured Payers	-0.5389	-0.9290	-0.3901
	(0.2048)	(0.1439)	

NOTES: TWFE is the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) from column (D) of the headline tables. Column (D) adds the state poverty rate, unemployment rate, log median household income, and an indicator for the two years before adoption to the preferred column (B), which has state, year, and region × year fixed effects. CSDID is the simple-aggregate average ATT from the Callaway and Sant'Anna (2021) doubly-robust group-time estimator, with states not expanded by 2019 as the comparison group and the same three time-varying covariates. Standard errors (in parentheses) are clustered at the state level.

A.3 Survey Source and Comparison States

The paper's uninsured series comes from the Current Population Survey through 2012 and the American Community Survey from 2013. Because the survey changes the year before most states adopted, Table A.9 re-estimates every column on the same series for 2008–2019 and on an American Community Survey series for 2008–2019. The preferred estimate falls from a 29% decline to a 26% decline when the window is shortened and to a 24% decline on the American Community Survey alone (from 26% to 24% and 22% weighted), and the indicator for the two years before adoption is no longer significant in the American Community Survey series. Table A.10 shows that the post-adoption event-study coefficients are nearly the same on the two series over the same years.

Table A.11 reports three further checks. Dropping Wisconsin, which covered childless adults up to 100% of the federal poverty level from April 2014 under a waiver and is the only Great Lakes state that had not expanded by 2019, makes every estimate somewhat larger. Dropping the Far West, the Mideast, and New England, where every state had expanded by 2019 and the region-by-year comparison rests only on adoption timing, changes little. The balancing test regresses each covariate on the treatment variable, and only the weighted poverty rate responds, by 0.4 percentage points. Panel C shows that total asthma inpatient stays did not change, while Medicaid-paid stays rose and uninsured stays fell.

Table A.9: Log Uninsured Rate, Spliced CPS/ACS Series against ACS Only

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Panel A: Spliced CPS/ACS, 2001–2019 (as in the paper)							
Unweighted	-0.332***	-0.342***	-0.346***	-0.358***	-0.348***	-0.343***	-0.255***
	(0.031)	(0.038)	(0.043)	(0.045)	(0.046)	(0.044)	(0.033)
1–2y pre				-0.042*	-0.028		
				(0.024)	(0.025)		
Weighted by population	-0.333***	-0.300***	-0.313***	-0.331***	-0.320***	-0.315***	-0.214***
	(0.038)	(0.045)	(0.045)	(0.048)	(0.054)	(0.046)	(0.036)
1–2y pre				-0.063**	-0.033		
				(0.030)	(0.020)		
Observations	969	969	969	969	779	969	969
Panel B: Spliced CPS/ACS, 2008–2019							
Unweighted	-0.290***	-0.305***	-0.315***	-0.329***	-0.324***	-0.311***	-0.215***
	(0.029)	(0.036)	(0.040)	(0.043)	(0.047)	(0.040)	(0.030)
1–2y pre				-0.029	-0.014		
				(0.025)	(0.028)		
Weighted by population	-0.300***	-0.268***	-0.278***	-0.302***	-0.298***	-0.278***	-0.182***
	(0.040)	(0.041)	(0.044)	(0.048)	(0.054)	(0.045)	(0.032)
1–2y pre				-0.052*	-0.020		
				(0.031)	(0.022)		
Observations	612	612	612	612	492	612	612
Panel C: ACS only, 2008–2019							
Unweighted	-0.249***	-0.277***	-0.287***	-0.300***	-0.293***	-0.287***	-0.201***
	(0.029)	(0.035)	(0.037)	(0.039)	(0.043)	(0.037)	(0.031)
1–2y pre				-0.027	-0.013		
				(0.019)	(0.022)		
Weighted by population	-0.266***	-0.245***	-0.256***	-0.269***	-0.262***	-0.256***	-0.176***
	(0.043)	(0.038)	(0.040)	(0.042)	(0.046)	(0.040)	(0.034)
1–2y pre				-0.028	-0.005		
				(0.018)	(0.017)		
Observations	612	612	612	612	492	612	612
State FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Region × year		Y	Y	Y	Y	Y	Y
Poverty, unemployment, log median income			Y	Y	Y	Y	Y
1–2y pre indicator				Y	Y		
Excludes early-coverage states					Y		
Log Medicare enr.						Y	
State linear trends							Y

NOTES: Columns as in the paper's uninsured-rate table. Panel A repeats the paper's estimates, whose uninsured series comes from the CPS through 2012 and the ACS from 2013. Panel B uses the same series for 2008–2019 only. Panel C uses an ACS-only series for 2008–2019, the uninsured share of the civilian noninstitutionalized population from ACS 1-year table B27001. Standard errors clustered by state in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.10: Event-Study Coefficients for the Log Uninsured Rate, 2008–2019, Spliced Series against ACS Only

	Unweighted		Weighted	
Years from adoption	Spliced	ACS only	Spliced	ACS only
-5	0.047	0.006	0.028	0.017
	(0.030)	(0.021)	(0.022)	(0.019)
-4	0.049	0.027	0.042	0.030
	(0.042)	(0.025)	(0.032)	(0.023)
-3	0.032	0.010	0.046	0.020
	(0.041)	(0.024)	(0.038)	(0.025)
-2	0.025	-0.006	0.001	0.008
	(0.040)	(0.028)	(0.036)	(0.026)
-1	-0.008	-0.014	-0.022	-0.010
	(0.043)	(0.031)	(0.036)	(0.030)
0	-0.156***	-0.157***	-0.159***	-0.144***
	(0.052)	(0.041)	(0.049)	(0.039)
1	-0.277***	-0.274***	-0.271***	-0.253***
	(0.061)	(0.053)	(0.062)	(0.050)
2	-0.351***	-0.345***	-0.319***	-0.303***
	(0.064)	(0.056)	(0.070)	(0.057)
3	-0.355***	-0.345***	-0.324***	-0.304***
	(0.064)	(0.057)	(0.069)	(0.056)
4	-0.359***	-0.345***	-0.322***	-0.298***
	(0.066)	(0.057)	(0.071)	(0.059)
5+	-0.357***	-0.336***	-0.343***	-0.307***
	(0.065)	(0.057)	(0.073)	(0.065)
Joint p, leads	0.287	0.160	0.077	0.319

NOTES: Coefficients on indicators for years relative to adoption from a regression of the log uninsured rate on those indicators, state, year, and BEA region-by-year fixed effects, and the state poverty rate, unemployment rate, and log median household income, for 2008–2019. The reference category is six or more years before adoption, together with states that had not adopted by 2019. Standard errors clustered by state in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.11: Verification Checks, State Panel

	Uninsured rate	Medicaid enrollment	PHI enrollment	Medicaid ED visits	Self-pay ED visits	Total ED visits
Panel A: Sample checks, log outcomes						
Unweighted						
(a) Column (B)	-0.342***	0.348***	-0.027***	0.456***	-0.574***	0.004
	(0.038)	(0.041)	(0.009)	(0.082)	(0.199)	(0.025)
(b) Excluding Wisconsin	-0.367***	0.362***	-0.028***	0.504***	-0.646***	0.014
	(0.034)	(0.043)	(0.010)	(0.077)	(0.213)	(0.027)
(c) Excluding fully expanded regions	-0.331***	0.345***	-0.025**	0.400***	-0.554**	-0.022
	(0.042)	(0.046)	(0.010)	(0.081)	(0.238)	(0.014)
Weighted by population						
(a) Column (B)	-0.300***	0.280***	-0.030***	0.426***	-0.552**	-0.011
	(0.045)	(0.037)	(0.008)	(0.094)	(0.265)	(0.022)
(b) Excluding Wisconsin	-0.332***	0.292***	-0.033***	0.513***	-0.686**	0.003
	(0.040)	(0.041)	(0.008)	(0.072)	(0.288)	(0.026)
(c) Excluding fully expanded regions	-0.303***	0.279***	-0.030***	0.401***	-0.544*	-0.023
	(0.048)	(0.039)	(0.008)	(0.100)	(0.286)	(0.019)
Observations, (a)	969	969	969	403	403	403
Observations, (b)	950	950	950	389	389	389
Observations, (c)	627	627	627	263	263	263

Panel B: Balancing test

	Poverty rate	Unemployment rate	Log median income
Unweighted	0.208	0.236	-0.001
	(0.170)	(0.242)	(0.011)
Weighted by population	0.408**	0.299	-0.010
	(0.203)	(0.211)	(0.008)
Observations	969	969	969

Panel C: Log asthma inpatient stays per resident

	Total	Medicaid	Uninsured
Unweighted	-0.005	0.170*	-0.558**
	(0.057)	(0.094)	(0.221)
Weighted by population	0.003	0.144	-0.545**
	(0.046)	(0.101)	(0.200)
Observations	365	472	368

NOTES: Each cell is the coefficient on the Medicaid expansion treatment variable (the share of the year after expansion) from a separate state-year regression with state, year, and BEA region-by-year fixed effects, as in column (B) of the headline tables. Weighted rows weight by the state's average population. Row (a) repeats column (B). Row (b) drops Wisconsin, which covered childless adults up to 100 percent of the federal poverty level from April 2014 under a waiver and is the only Great Lakes state not expanded by 2019. Row (c) drops the three BEA regions in which every state expanded by 2019 (Far West, Mideast, and New England), so each remaining region has a comparison state. The ED outcomes cover 403 state-years in 36 states, 2006–2019, without Illinois. Panel B regresses each covariate on the treatment variable, and the poverty and unemployment rates are in percentage points. Panel C uses asthma inpatient stays per resident, 2003–2017, for Medicaid, the uninsured, and the total over the four payers reported (Medicaid, Medicare, private, and uninsured), which is missing when any payer is missing. For outcomes in logs, the percent effect of a coefficient b is $100(e^b - 1)$. Standard errors clustered by state in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

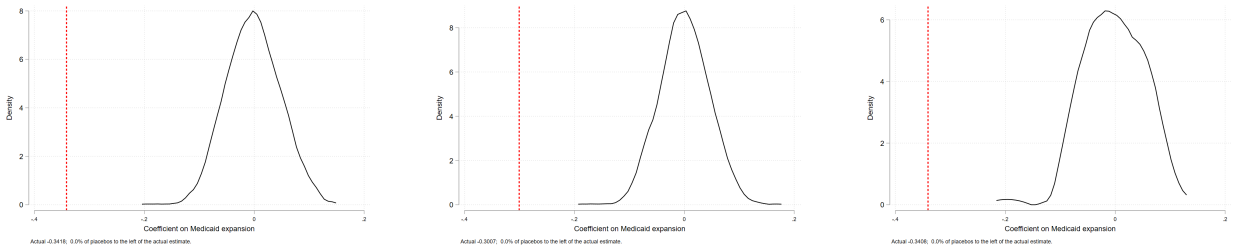
A.4 Randomization Inference

Standard errors clustered by state can be unreliable when the number of treated or control clusters is modest. As an alternative, I ask how unusual the actual estimate is compared with estimates from placebo assignments of treatment. In each of 1,000 draws (100 for the negative binomial model), I randomly reassign the actual list of adoption years to states without replacement, which preserves the cohort structure, shift the whole block of adoption dates by a common number of years within the 2001–2019 window, and re-estimate the preferred specification. Figure A.1 through Figure A.3 plot the resulting distributions; the red line marks the actual estimate, and the note beneath each panel reports the share of placebo estimates more extreme than that estimate.

For the uninsured rate and for self-pay ED visits, none of the 1,000 unweighted placebo estimates are as negative as the actual estimate. For PHI enrollment, whose effect is small in absolute terms, 0.4 percent of placebo estimates are more negative than the actual estimate. The placebo distributions are centered on zero in every case, which indicates that the preferred specification does not mechanically produce effects of the kind estimated.

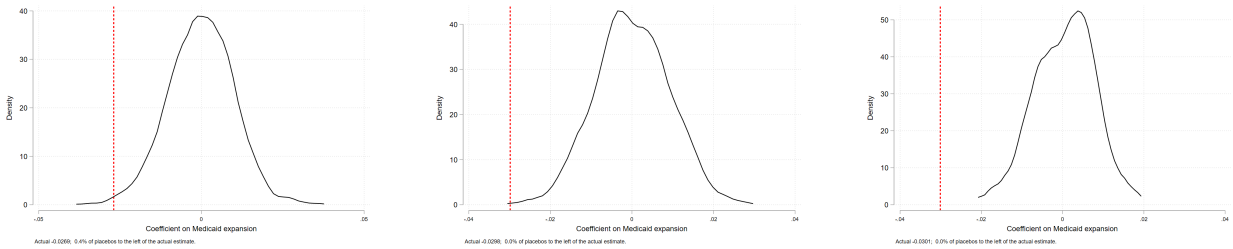
Figure A.1: Randomization Inference for Coverage Outcomes

(a) Log uninsured rate



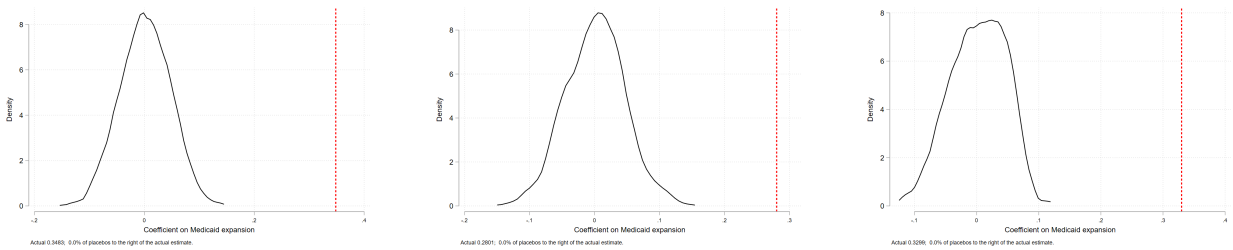
Unweighted OLS **Weighted OLS** **Negative binomial**

(b) Log PHI enrollment rate



Unweighted OLS **Weighted OLS** **Negative binomial**

(c) Log Medicaid enrollment rate

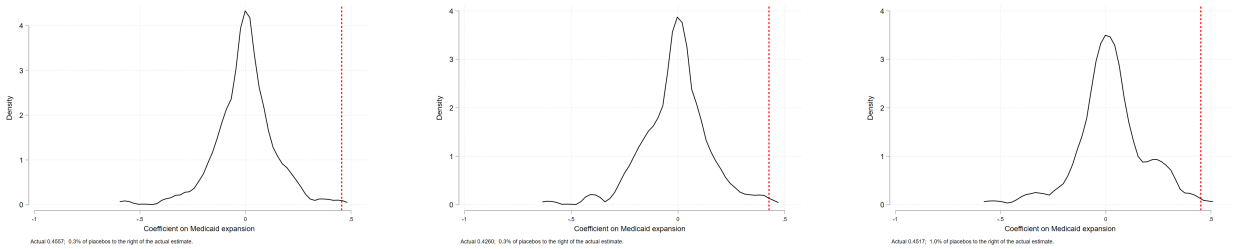


Unweighted OLS **Weighted OLS** **Negative binomial**

NOTES: Each panel shows the distribution of the coefficient on the Medicaid-expansion indicator across 1,000 placebo draws (100 for the negative binomial). Each draw randomly reassigns the actual list of adoption years to states without replacement, preserving the cohort structure, and shifts the block of adoption dates by a common integer offset within the 2001–2019 window; the preferred specification is then re-estimated on the placebo treatment. The red vertical line marks the actual estimate.

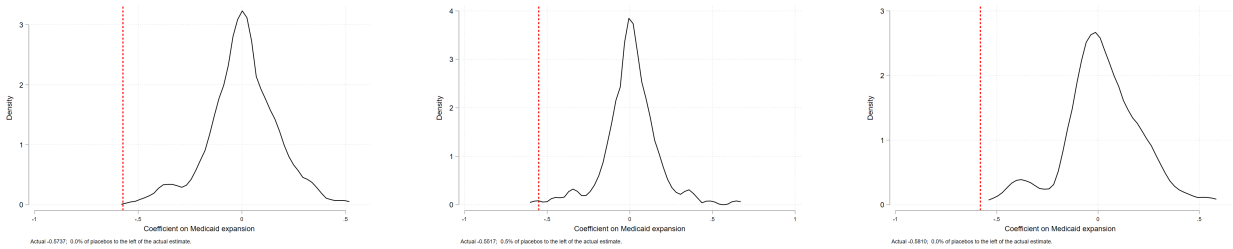
Figure A.2: Randomization Inference for ED Visits by Payer

(a) Log Medicaid ED visits



Unweighted OLS **Weighted OLS** **Negative binomial**

(b) Log self-pay/no-charge ED visits

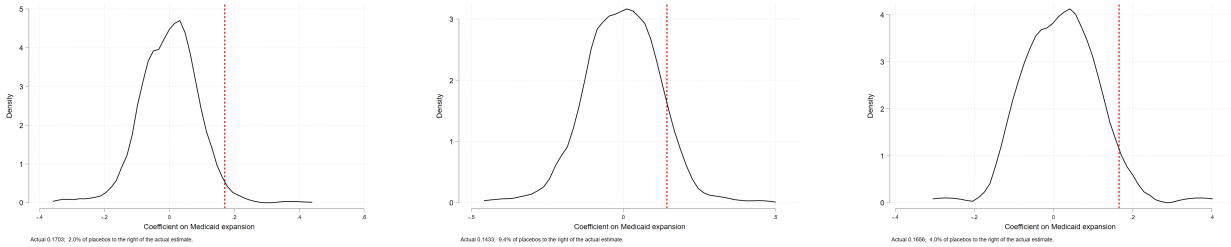


Unweighted OLS **Weighted OLS** **Negative binomial**

NOTES: As in Figure A.1. ED samples exclude Illinois.

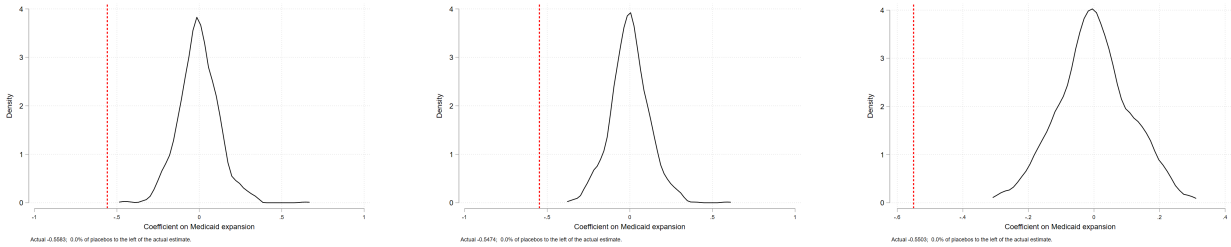
Figure A.3: Randomization Inference for Asthma Inpatient Stays

(a) Log asthma inpatient stays, Medicaid payers



Unweighted OLS **Weighted OLS** **Negative binomial**

(b) Log asthma inpatient stays, uninsured payers



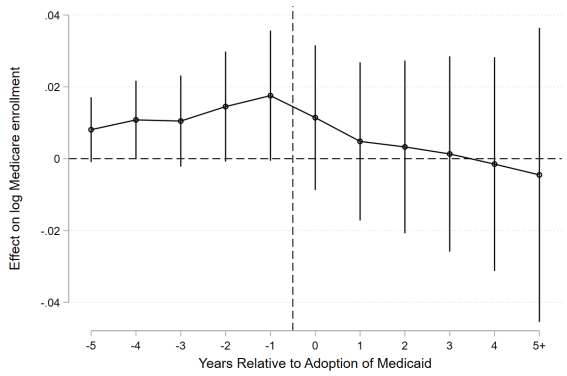
Unweighted OLS **Weighted OLS** **Negative binomial**

NOTES: As in Figure A.1. The sample covers 2003–2017 and excludes Illinois.

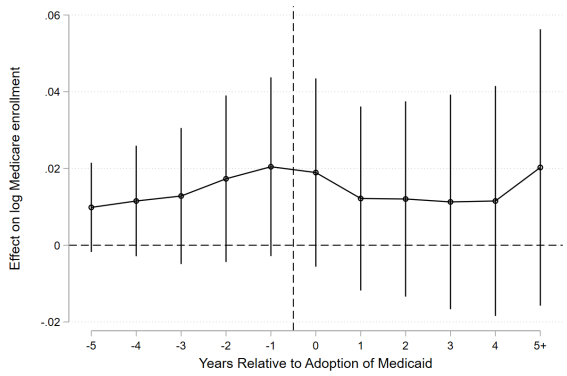
A.5 Additional Event Studies

Figure A.4 plots the event study for Medicare enrollment, the placebo outcome. The coefficients are small throughout, between about 1 and 2% before adoption and between −0.5 and 2% after, and none is significant at the 5% level; this is small next to the effects on the uninsured rate and on Medicaid enrollment, which are on the order of 30 to 40%. The figure also shows total ED visits, which should not move if expansion changed who paid rather than how much care was delivered. Figure A.5 shows the event studies for the remaining coverage outcomes, and Figure A.6 shows the negative binomial versions of the uninsured and ED event studies. In every case the reference category is six or more years before adoption, together with states that had not adopted by 2019.

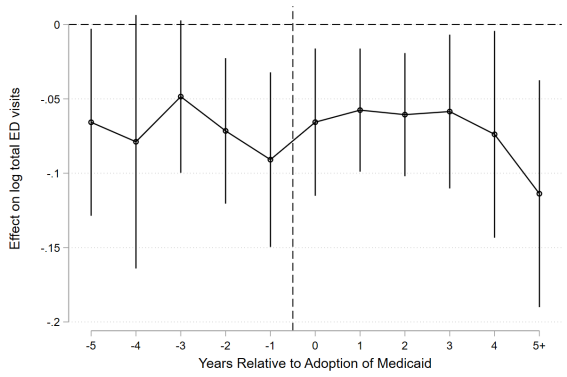
Figure A.4: Event Studies for Placebo and Total Outcomes



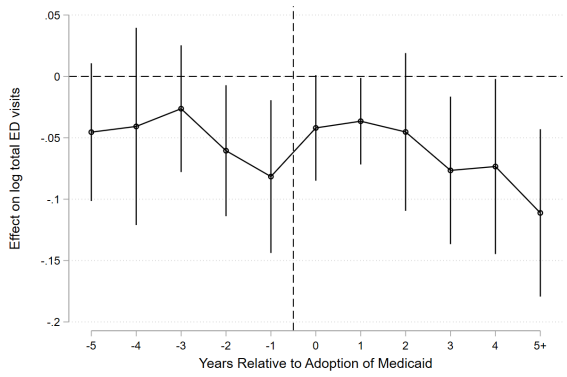
(a) Log Medicare enrollment, unweighted



(b) Log Medicare enrollment, weighted



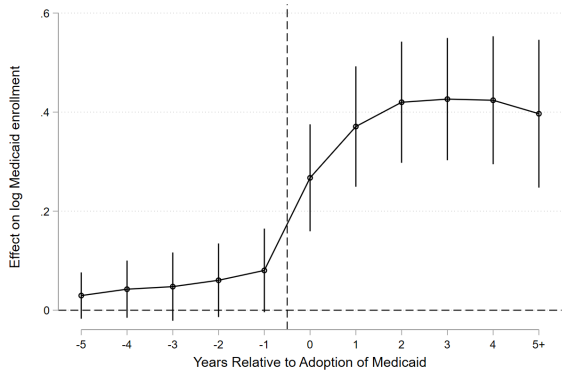
(c) Log total ED visits, unweighted



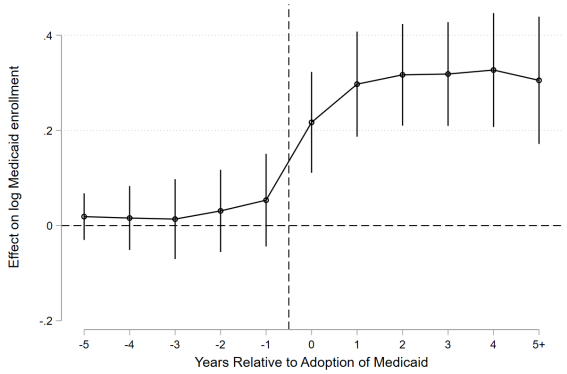
(d) Log total ED visits, weighted

NOTES: Each panel plots coefficients on indicators for years relative to adoption from a regression that also includes state fixed effects, year fixed effects, BEA region-by-year fixed effects, and the state poverty rate, unemployment rate, and log median household income. The reference category is six or more years before adoption, together with states that had not adopted by 2019. Vertical bars are 95% confidence intervals from standard errors clustered by state. ED samples exclude Illinois.

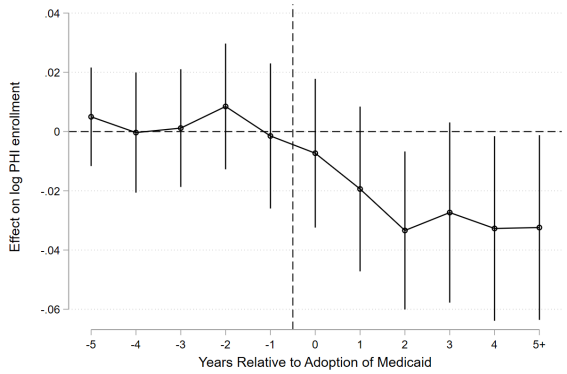
Figure A.5: Event Studies for Medicaid and Private Enrollment



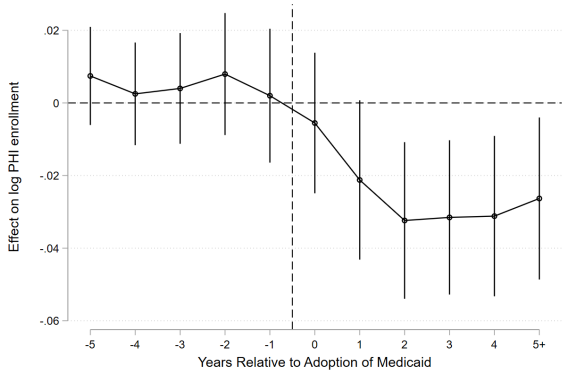
(a) Log Medicaid enrollment, unweighted



(b) Log Medicaid enrollment, weighted



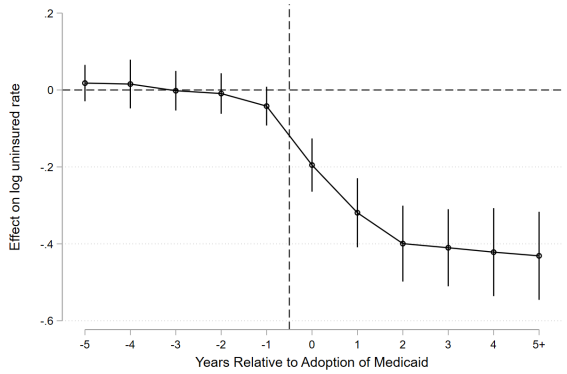
(c) Log PHI enrollment, unweighted



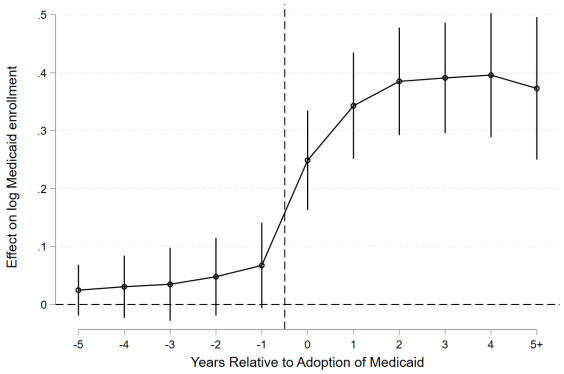
(d) Log PHI enrollment, weighted

NOTES: As in Figure A.4.

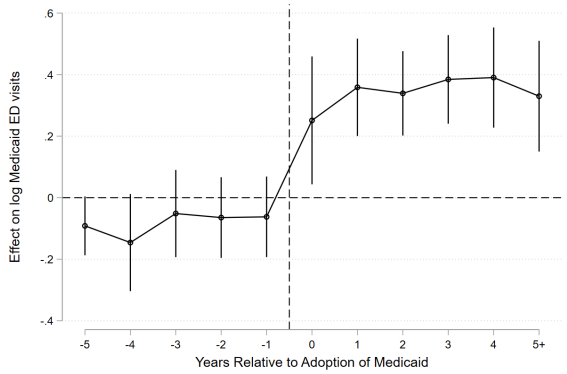
Figure A.6: Event Studies from Negative Binomial Models



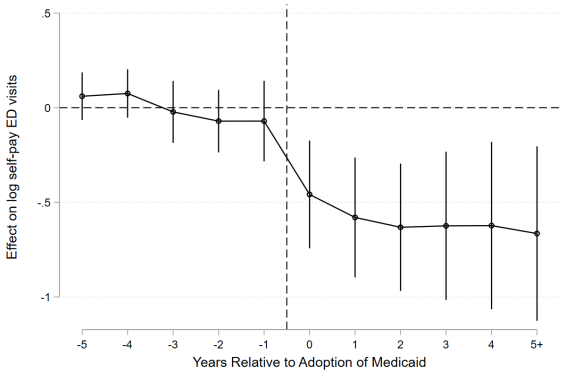
(a) Uninsured count



(b) Medicaid enrollee count



(c) Medicaid ED visits



(d) Self-pay/no-charge ED visits

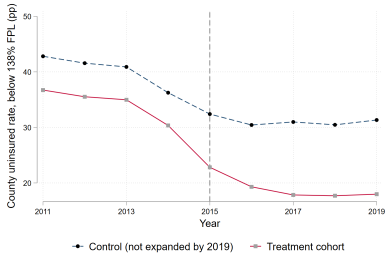
NOTES: As in Figure A.4, except each panel is a negative binomial regression of the count on the same covariates with state population as the exposure, so coefficients are log incidence-rate ratios.

B Rural Hospital Design

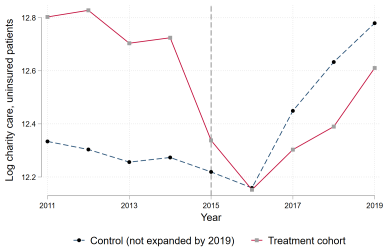
B.1 Raw Trends for Later Adoption Cohorts

The paper plots raw means for the 2014 cohort. Figure B.1 through Figure B.4 repeat the comparison for the cohorts that adopted in 2015 (Alaska, Indiana, and Pennsylvania), 2016 (Louisiana and Montana), 2018 (Maine), and 2019 (Virginia), each against rural hospitals in states that had not expanded by 2019. The later cohorts contain few hospitals, so their series are noisier, but they offer more pre-adoption years.

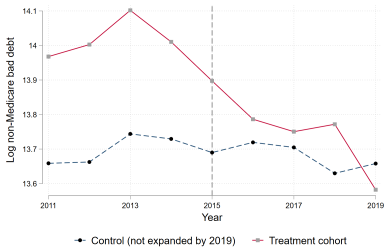
Figure B.1: Rural Hospital Outcomes Before and After Adoption, 2015 Cohort (Alaska, Indiana, Pennsylvania)



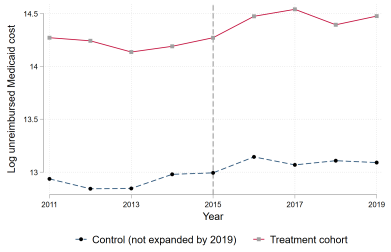
(a) County uninsured, below 138% FPL



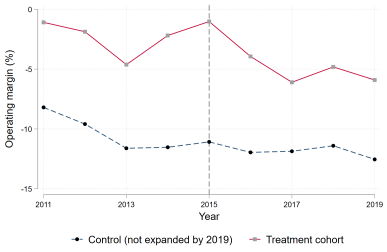
(b) Log charity care, uninsured



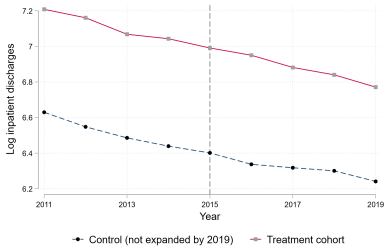
(c) Log non-Medicare bad debt



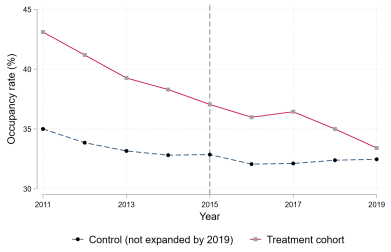
(d) Log unreimbursed Medicaid cost



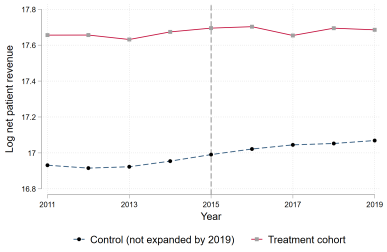
(e) Operating margin



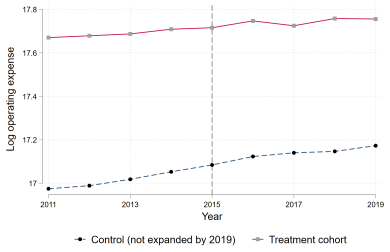
(f) Log inpatient discharges



(g) Occupancy rate



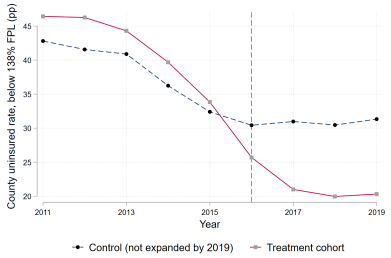
(h) Log net patient revenue



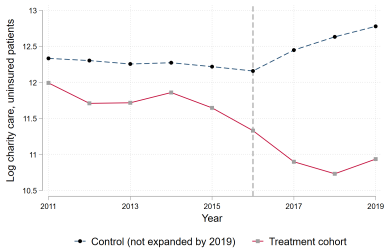
(i) Log operating expense

NOTES: As in the paper's 2014-cohort figure, each panel plots means across rural hospitals in the cohort and in states that had not expanded by 2019. The vertical dashed line marks the cohort's adoption year.

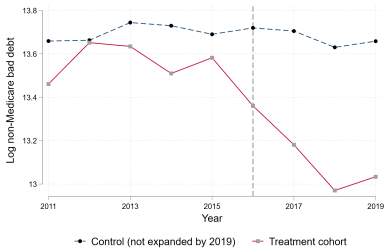
Figure B.2: Rural Hospital Outcomes Before and After Adoption, 2016 Cohort (Louisiana, Montana)



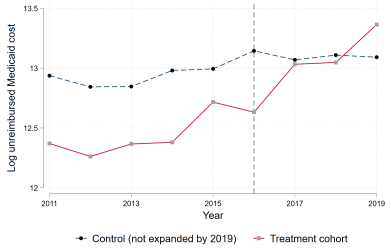
(a) County uninsured, below 138% FPL



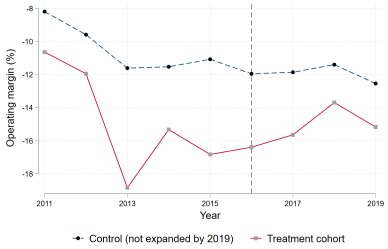
(b) Log charity care, uninsured



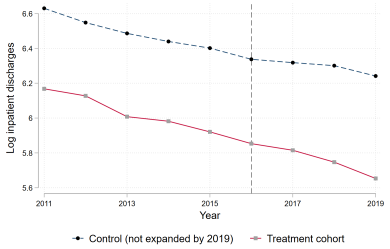
(c) Log non-Medicare bad debt



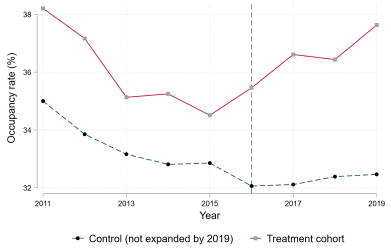
(d) Log unreimbursed Medicaid cost



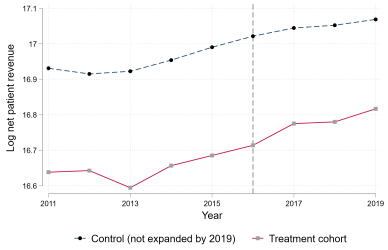
(e) Operating margin



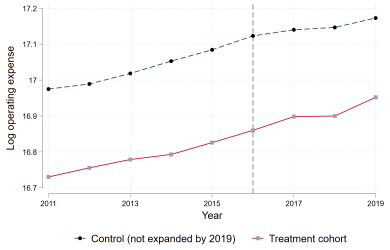
(f) Log inpatient discharges



(g) Occupancy rate



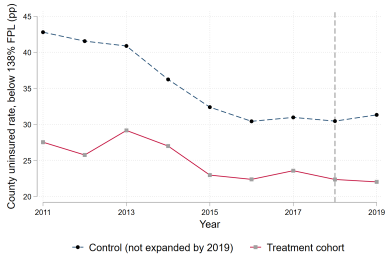
(h) Log net patient revenue



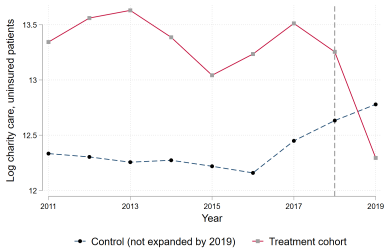
(i) Log operating expense

NOTES: As in the paper's 2014-cohort figure, each panel plots means across rural hospitals in the cohort and in states that had not expanded by 2019. The vertical dashed line marks the cohort's adoption year.

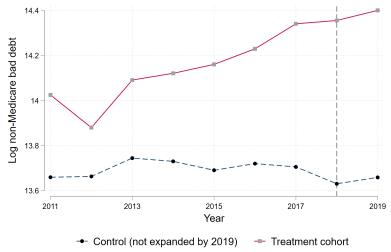
Figure B.3: Rural Hospital Outcomes Before and After Adoption, 2018 Cohort (Maine)



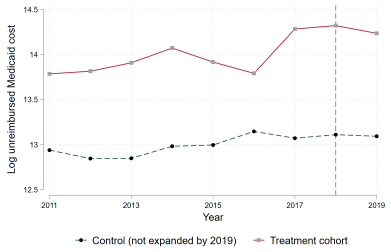
(a) County uninsured, below 138% FPL



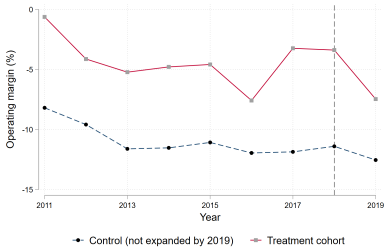
(b) Log charity care, uninsured



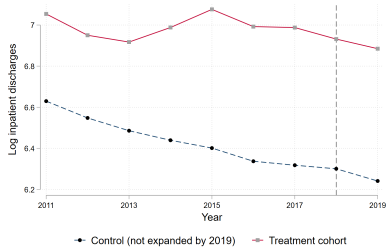
(c) Log non-Medicare bad debt



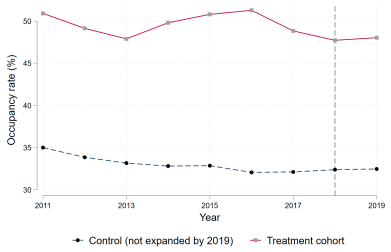
(d) Log unreimbursed Medicaid cost



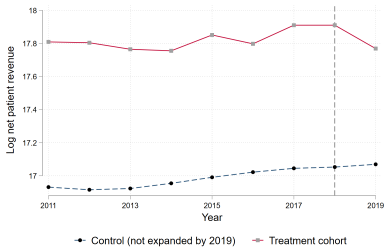
(e) Operating margin



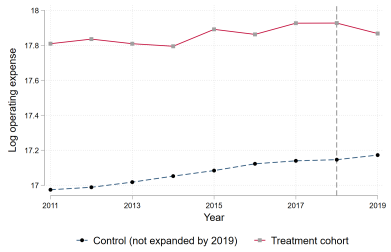
(f) Log inpatient discharges



(g) Occupancy rate



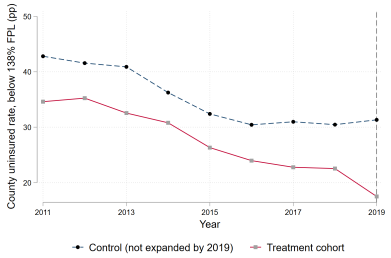
(h) Log net patient revenue



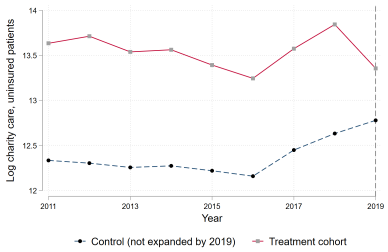
(i) Log operating expense

NOTES: As in the paper's 2014-cohort figure, each panel plots means across rural hospitals in the cohort and in states that had not expanded by 2019. The vertical dashed line marks the cohort's adoption year.

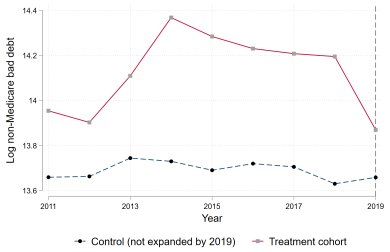
Figure B.4: Rural Hospital Outcomes Before and After Adoption, 2019 Cohort (Virginia)



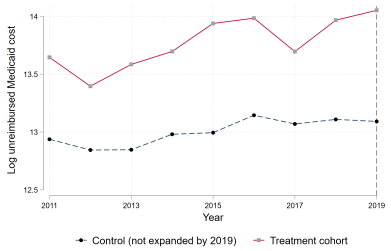
(a) County uninsured, below 138% FPL



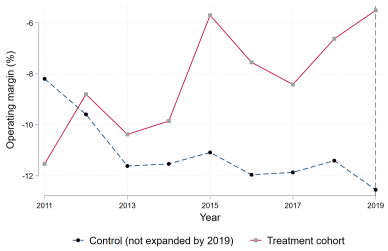
(b) Log charity care, uninsured



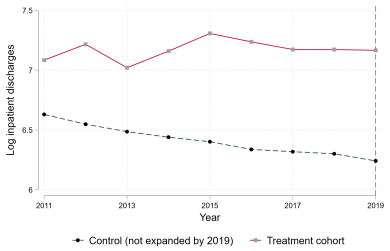
(c) Log non-Medicare bad debt



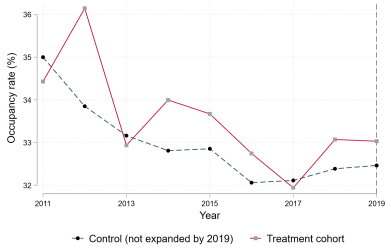
(d) Log unreimbursed Medicaid cost



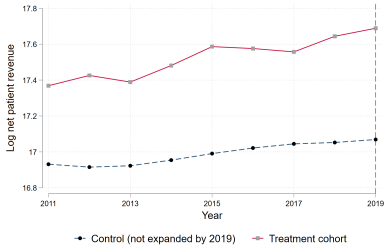
(e) Operating margin



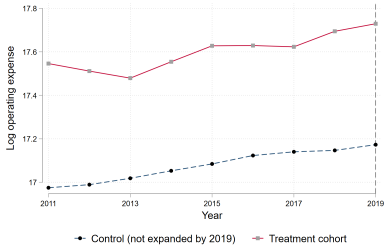
(f) Log inpatient discharges



(g) Occupancy rate



(h) Log net patient revenue



(i) Log operating expense

NOTES: As in the paper's 2014-cohort figure, each panel plots means across rural hospitals in the cohort and in states that had not expanded by 2019. The vertical dashed line marks the cohort's adoption year.

B.2 Pre-Trend Diagnostics

The paper measures charity care, bad debt, and unreimbursed Medicaid cost in logs of 2019 dollars, and it measures treatment by the share of each hospital's cost-reporting period that falls after its state's expansion date. Only about a third of rural hospitals have fiscal years that begin in January. Under a simple indicator for years at or after expansion, the reference year before adoption already contains post-expansion months for 504 of 1,207 treated hospital-years, and charity care as a share of operating expense then fails the pre-adoption test ($p = 0.01$). With the share of the reporting period, every payer-mix outcome passes the test, as a share and in logs. Operating margin passes it as well ($p = 0.23$), but its earliest pooled coefficient is about as large as the estimated effect, and occupancy fails it narrowly ($p = 0.047$). This subsection asks whether covariates or trend adjustments change the estimates, where the remaining pre-trends come from, and how much they matter.

Table B.1 re-estimates every hospital outcome on one common sample under seven specifications. S0 is column 3 of the main tables, the preferred specification plus the county covariates. S1 adds county demographics from the American Community Survey (the share with a bachelor's degree, median age, the shares white and Black, log population, and the share aged 65 or older). S2 instead adds year effects interacted with each baseline hospital characteristic, so that hospitals of each ownership type, critical access status, size, teaching status, DSH status, and sole-hospital status can follow their own path. S3 combines the two, and S4 adds year effects interacted with the quintile of the hospital's 2011 value of the outcome, to allow for mean reversion. S5 adds state linear trends, and S6 estimates state linear trends only on observations not yet treated and removes them from every year, so the trend is not fit to the response.

The payer-mix estimates barely move across these specifications. In logs, the charity care decline ranges from 56 to 61% and the bad debt decline from 21 to 28%, and the Medicaid shortfall rise ranges from 15 to 26%, though it is imprecise under S6. As shares of operating expense the pattern is the same, and no specification rejects a flat pre-period for the charity care share at the 5% level. Table B.2 shows the charity care share test by group. It passes in the pooled sample ($p = 0.13$) but rejects for government hospitals ($p = 0.02$) and, to a lesser degree, for-profit hospitals ($p = 0.06$), and not for nonprofit hospitals, critical access hospitals, or the 2014 cohort.

The operating margin estimate ranges from 1.96 to 2.12 points under S0 to S4 and rises under the two trend adjustments, and its pre-period test is passed in every specification except S5. Table B.2 shows where its large earliest pre-adoption coefficient comes from. It is concentrated among government hospitals ($p < 0.001$), which are 41% of rural hospitals in non-adopting states and 28% in adopting states, and among critical access hospitals ($p = 0.02$), and it is absent in the 2014 cohort ($p = 0.34$), whose margin rises by 2.36 points (standard error 0.80). The earliest event-study coefficient pools four or more years before adoption and draws on the 2015, 2016, 2018, and 2019 cohorts.

Table B.3 reports robust confidence intervals for the average post-adoption effect following Rambachan and Roth (2023). These allow post-adoption departures from parallel trends of up

to $\bar{M}$ times the largest change between adjacent pre-adoption periods. Log charity care remains significantly negative up to $\bar{M} = 0.8$, and up to at least 2 when the pooled earliest period is left out. The share version holds up to 0.6, and up to 1.5 without that period. The operating margin interval includes zero at any $\bar{M}$ above zero in S0, and at any $\bar{M}$ above 0.3 without the pooled earliest period, which is why the paper treats the margin estimate as suggestive. Figure B.5 plots the event studies for the charity care share and operating margin under S0, S3, and S6.

Table B.1: Rural Hospitals: Average Effects and Pre-Period Tests under Covariate and Trend Adjustments

	(S0) Column 3	(S1) + County demographics	(S2) + Baseline type × year	(S3) S1 + S2	(S4) S3 + 2011 outcome × year	(S5) State trends	(S6) Pre-period state trends
County uninsured rate, adults <138% FPL	-7.968***	-7.953***	-7.956***	-7.941***	-8.598***	-6.691***	-7.395***
	(1.128)	(1.127)	(1.079)	(1.079)	(1.073)	(1.046)	(1.433)
Pre-period p	[0.220]	[0.203]	[0.247]	[0.231]	[0.293]	[0.030]	[0.243]
Charity care, uninsured patients	-1.297***	-1.293***	-1.281***	-1.276***	-1.285***	-0.883***	-1.106***
	(0.283)	(0.282)	(0.265)	(0.264)	(0.262)	(0.250)	(0.255)
Pre-period p	[0.133]	[0.137]	[0.081]	[0.082]	[0.115]	[0.694]	[0.231]
Non-Medicare bad debt	-1.004***	-1.011***	-1.016***	-1.024***	-1.102***	-0.943***	-1.281***
	(0.252)	(0.249)	(0.252)	(0.249)	(0.233)	(0.267)	(0.344)
Pre-period p	[0.732]	[0.643]	[0.879]	[0.813]	[0.464]	[0.407]	[0.171]
Unreimbursed Medicaid cost	0.754***	0.756***	0.756***	0.758***	0.804***	0.524**	0.807*
	(0.203)	(0.206)	(0.223)	(0.225)	(0.196)	(0.206)	(0.467)
Pre-period p	[0.965]	[0.963]	[0.979]	[0.981]	[0.612]	[0.110]	[0.275]
Log charity care, uninsured patients	-0.898***	-0.894***	-0.914***	-0.910***	-0.917***	-0.821***	-0.943***
	(0.155)	(0.153)	(0.152)	(0.151)	(0.145)	(0.113)	(0.248)
Pre-period p	[0.312]	[0.261]	[0.444]	[0.387]	[0.365]	[0.600]	[0.824]
Log non-Medicare bad debt	-0.325***	-0.324***	-0.327***	-0.327***	-0.307***	-0.237***	-0.326***
	(0.045)	(0.045)	(0.047)	(0.047)	(0.046)	(0.051)	(0.095)
Pre-period p	[0.717]	[0.717]	[0.747]	[0.736]	[0.809]	[0.411]	[0.586]
Log unreimbursed Medicaid cost	0.176*	0.180*	0.175*	0.179*	0.233***	0.142***	0.212
	(0.098)	(0.099)	(0.100)	(0.100)	(0.090)	(0.054)	(0.240)
Pre-period p	[0.850]	[0.885]	[0.887]	[0.920]	[0.476]	[0.004]	[0.375]
Operating margin	2.121***	2.116***	2.100***	2.100***	1.957**	3.565***	4.770***
	(0.731)	(0.716)	(0.733)	(0.724)	(0.799)	(0.728)	(1.655)
Pre-period p	[0.222]	[0.259]	[0.184]	[0.220]	[0.133]	[0.001]	[0.126]
Log inpatient discharges	-0.030	-0.031*	-0.027	-0.028	-0.029	0.025	0.013
	(0.020)	(0.019)	(0.022)	(0.022)	(0.022)	(0.017)	(0.023)
Pre-period p	[0.149]	[0.131]	[0.143]	[0.126]	[0.151]	[0.357]	[0.690]
Occupancy rate	-0.393	-0.360	-0.363	-0.324	-0.205	0.589	0.274
	(0.538)	(0.531)	(0.581)	(0.576)	(0.603)	(0.508)	(0.740)
Pre-period p	[0.042]	[0.050]	[0.181]	[0.206]	[0.538]	[0.017]	[0.039]
Log net patient revenue	0.013	0.013	0.011	0.011	0.016	0.028***	0.084***
	(0.014)	(0.014)	(0.013)	(0.013)	(0.013)	(0.010)	(0.022)
Pre-period p	[0.190]	[0.214]	[0.250]	[0.281]	[0.437]	[0.107]	[0.625]
Log operating expense	-0.003	-0.003	-0.003	-0.003	0.001	-0.004	-0.014
	(0.011)	(0.011)	(0.009)	(0.009)	(0.010)	(0.008)	(0.036)
Pre-period p	[0.063]	[0.071]	[0.051]	[0.057]	[0.124]	[0.044]	[0.505]

NOTES: Each cell is the coefficient on Post, the share of the hospital's cost-reporting period after its state's expansion date (for the county uninsured rate, the share of the calendar year), with its standard error, from a separate regression on one common sample of rural hospitals, 2011–2019 (hospital-years with every covariate and baseline characteristic observed). The payer-mix outcomes appear as shares of operating expense, in percent, and in logs of 2019 dollars. The bracketed number is the p-value for the joint test that the three pre-adoption coefficients (four or more, three, and two years before) are zero in the matching event study. S0 is the preferred specification (hospital and region-by-year fixed effects and county poverty, unemployment, and log median income). S1 adds the county share with a bachelor's degree, median age, shares white and Black, log population, and share aged 65 or older (ACS 5-year). S2 adds year fixed effects interacted with each baseline characteristic (ownership, critical access status, size, teaching status, DSH status, and sole hospital in the county, all fixed at 2011–2013). S4 adds year fixed effects interacted with the quintile of the hospital's 2011 value of the outcome. S5 adds state linear trends. S6 estimates state linear trends on observations not yet treated, removes them from the outcome in every year, and re-estimates S0; its standard errors ignore the first step. Standard errors clustered by state. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.2: Pre-Adoption Event-Study Coefficients by Hospital Group

	Charity care, uninsured					Operating margin				
	≤ -4	-3	-2	Joint p	DiD	≤ -4	-3	-2	Joint p	DiD
Nonprofit	0.182	0.072	0.119	0.521	-1.343 (0.308)	1.106	1.056	0.175	0.580	1.121 (0.848)
For-profit	1.285	0.604	0.375	0.061	-2.077 (0.653)	3.279	0.245	0.119	0.324	6.390 (1.960)
Government	0.517	0.012	0.031	0.024	-1.042 (0.322)	4.849	-0.640	0.153	0.000	2.408 (0.887)
Critical access	0.252	0.081	0.096	0.273	-1.089 (0.281)	2.545	0.135	0.120	0.023	1.853 (0.924)
Not critical access	0.499	0.084	0.106	0.101	-1.556 (0.248)	2.802	1.206	0.690	0.538	2.427 (1.146)
Excluding Southeast region	0.258	0.064	-0.004	0.587	-0.693 (0.180)	2.316	0.940	0.108	0.354	2.215 (0.905)
2014 cohort and controls only	–	0.133	0.111	0.195	-1.232 (0.332)	–	0.687	0.610	0.338	2.359 (0.796)
All (S0, full sample)	0.349	0.093	0.104	0.130	-1.297 (0.283)	2.568	0.548	0.265	0.228	2.172 (0.718)
Nonprofit, not critical access	0.404	-0.039	0.094	0.241	-1.614 (0.332)	0.627	1.643	0.734	0.754	2.011 (1.503)

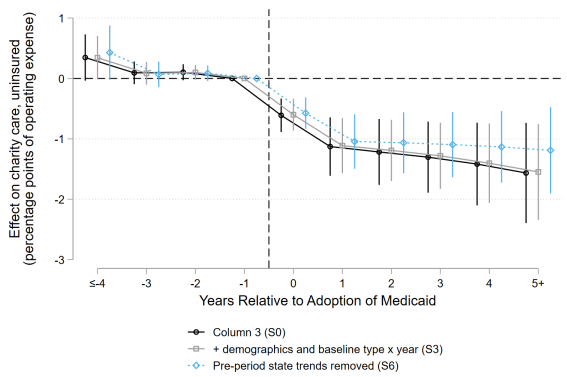
NOTES: Pre-adoption coefficients, relative to the year before adoption, from the preferred event study (hospital and region-by-year fixed effects, county covariates) estimated within each group of rural hospitals, 2011–2019. Charity care is in percent of operating expense and operating margin in percent. The 2014 cohort is observed only three and two years before adoption, so its joint test covers those two coefficients. DiD is the coefficient on Post, the share of the hospital's cost-reporting period after its state's expansion date, in the matching average-effect regression (standard error clustered by state in parentheses).

Table B.3: Sensitivity of the Average Post-Adoption Effect to Violations of Parallel Trends

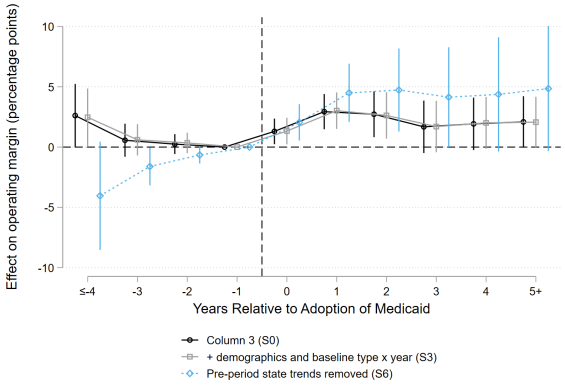
Outcome	Specification	Average effect	Breakdown $\bar{M}$	Robust CI, $\bar{M} = 0.5$	Robust CI, $\bar{M} = 1$
Charity care, uninsured patients	Column 3 (S0)	-1.207 (0.270)	0.6	[-2.272, -0.211]	[-3.034, 0.519]
	S3	-1.189 (0.255)	0.6	[-2.222, -0.245]	[-2.979, 0.480]
	S0, pre-periods -3, -2 only	-1.207 (0.270)	1.5	[-2.001, -0.547]	[-2.327, -0.295]
Log charity care, uninsured patients	Column 3 (S0)	-0.842 (0.148)	0.8	[-1.496, -0.315]	[-1.975, 0.129]
	S3	-0.852 (0.143)	0.8	[-1.488, -0.327]	[-1.954, 0.119]
	S0, pre-periods -3, -2 only	-0.842 (0.148)	≥ 2	[-1.172, -0.499]	[-1.262, -0.421]
Operating margin	Column 3 (S0)	2.109 (0.788)	0.0	[-4.598, 8.971]	[-10.986, 15.388]
	S3	2.125 (0.791)	0.1	[-4.035, 8.438]	[-9.842, 14.271]
	S0, pre-periods -3, -2 only	2.109 (0.788)	0.3	[-0.540, 4.824]	[-2.525, 6.811]

NOTES: The average effect is the mean of the six post-adoption event-study coefficients (standard error in parentheses), on the common sample of Table B.1. Robust 95% confidence intervals follow Rambachan and Roth (2023) under relative-magnitude restrictions, which allow post-adoption violations of parallel trends of up to $\bar{M}$ times the largest change between consecutive pre-adoption periods. The breakdown value is the largest $\bar{M}$ on a grid of 0 to 2 in steps of 0.1 at which the robust interval excludes zero. The earliest pre-period pools four or more years before adoption (identified by the 2015, 2016, 2018, and 2019 cohorts) and is treated as one period; the last row for each outcome drops it. S3 adds county demographics and baseline hospital type by year fixed effects.

Figure B.5: Event Studies under Covariate and Trend Adjustments



(a) Charity care for uninsured patients, % of operating expense



(b) Operating margin

NOTES: Event studies on the common sample of Table B.1, under the preferred specification (S0), with county demographics and baseline hospital type by year fixed effects (S3), and with state trends estimated before adoption removed (S6). Event time is counted from the hospital's first cost report that includes months after its state's expansion, and the reference year is the one before it. Vertical bars are 95% confidence intervals based on standard errors clustered by state.

B.3 Within-State Exposure Design

The hospital design in the paper treats every hospital in an adopting state as equally affected by expansion. Medicaid expansion, though, reaches a hospital through a more local channel, since a hospital treating many uninsured patients before 2014 had more to gain than one that was not. If expansion relieved hospitals, the relief should be greater where the burden was greater. In the spirit of Finkelstein (2007), I therefore also use variation in exposure within a state. I measure a hospital's exposure as its pre-expansion uncompensated care burden, the cost of uncompensated care as a share of operating expense, averaged over 2011–2013.[1] For hospital h in state s and year t, I estimate

$$Outcome_{hst} = \beta_0 \cdot Post_{hst} + \beta_1 \cdot (Z_h \times Post_{hst}) + \beta_2 \cdot X_{ct} + c_h + W_{st} + \sum_t \lambda_t \cdot (Z_h \times \mathbb{1}[year = t]) + \epsilon_{hst} \quad (1)$$

where Z_h is the hospital's standardized exposure, $Post_{hst}$ is the share of the hospital's cost-reporting period after state s's expansion date, as in the paper, X_{ct} contains the county poverty rate, unemployment rate, and log median household income, c_h are hospital fixed effects, and W_{st} are state-by-year fixed effects. The state-by-year fixed effects absorb everything common to a state in a year. For example, a state's Medicaid payment rates (its supplemental payments to hospitals that treat many low-income patients) and budget conditions are all absorbed. Because fiscal years differ across hospitals within a state, $Post_{hst}$ varies within a state and year, so it also enters on its own. The terms λ_t interact exposure with each calendar year, absorbing any national change in the relationship between exposure and outcomes; for example, if high-burden hospitals everywhere were on a different trajectory after 2014, that would alter λ_t (and not β_1). The coefficient β_1 therefore compares a high-exposure hospital with a low-exposure hospital in the same state and year, asking whether the gap between them changed after the state adopted relative to the same gap in states that did not. The identifying assumption is that the gap between high- and low-exposure hospitals within a state would have trended similarly in adopting and non-adopting states. Standard errors are clustered by state.

The estimates are in column 1 of Table B.4 and Table B.5, for the same rural hospitals, outcomes, and 2011–2019 window as the paper. Declines in charity care for the uninsured were larger at hospitals that had carried more uncompensated care before expansion, and a hospital one standard deviation higher in pre-period burden saw an additional decline of 1.74 percentage points, significant at the 1 percent level. The same hospitals saw their unreimbursed Medicaid cost rise by an additional 0.98 points per standard deviation, also significant at the 1 percent level, so the hospitals that gained the most newly insured patients also took on the most below-cost Medicaid business. Bad debt and the county uninsured rate show no gradient.

The operating margin gradient is −2.30 points per standard deviation, significant at the 5 percent

[1] To limit noise from hospitals with few pre-period reports, I shrink each hospital's average toward the mean of hospitals in the same state with the same critical-access status. I then standardize the measure on the 2013 cross-section of rural hospitals, so coefficients are per standard deviation of exposure.

level. Hospitals with heavier pre-expansion burdens therefore gained less in margins than other hospitals in their state, even though they lost more charity care. Their larger Medicaid shortfalls offset only part of that relief, so the payer-mix shift does not account for the margin gradient, which is another reason the paper treats the margin estimate as suggestive. Volume, occupancy, net patient revenue, and operating expense show no gradient.

Figure B.6 plots event-study versions of Equation 1 for four outcomes. Because the regression also controls for event time itself, every coefficient is an exposure gradient relative to the year before adoption. For charity care, the gradient is small before adoption, falls by about 1 point per standard deviation in the adoption year, and stays between −1.6 and −1.8 thereafter. The unreimbursed Medicaid cost gradient is slightly below zero before adoption, though not significantly, and it is near zero in the adoption year and 0.8 to 1.1 points higher in every later year. The operating margin gradient is imprecise before adoption, with one pre-adoption coefficient as negative as the post-adoption ones, and falls by about 1.5 to 3 points after, and the discharge gradient differs from zero only in the earliest pooled pre-period bin and does not change at adoption.

Table B.4: Within-State Exposure Design: Payer Mix

	(1)	(2)
County uninsured rate, adults <138% FPL	0.052	0.041
	(0.134)	(0.281)
Charity care, uninsured patients	-1.735***	-0.272*
	(0.359)	(0.153)
Non-Medicare bad debt	0.069	-0.403**
	(0.171)	(0.184)
Unreimbursed Medicaid cost	0.983***	0.268
	(0.247)	(0.230)
Exposure measure	Hospital burden	County eligibility
State × year FE	Y	Y
Exposure × year	Y	Y
County covariates	Y	Y

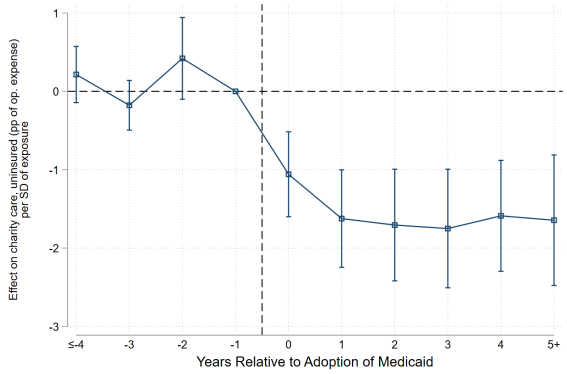
NOTES: Rural hospitals, 2011–2019, outcomes as in the main tables except that the payer-mix outcomes are shares of operating expense, in percent, as in the descriptive table. Each cell is the coefficient on the interaction of Post, defined as in the main tables, with a hospital's standardized exposure, so it is the change per standard deviation, from a separate regression of the row outcome. In column 1 exposure is the hospital's 2011–2013 uncompensated care share of operating expense, shrunk toward the mean of hospitals in the same state with the same critical access status; in column 2 it is the 2011–2013 share of the county's adults who were uninsured and below 138% of the federal poverty line. Every regression includes Post itself, hospital fixed effects, state-by-year fixed effects, interactions of exposure with each year, and the county covariates. Standard errors clustered by state in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.5: Within-State Exposure Design: Margins, Volume, and Cost

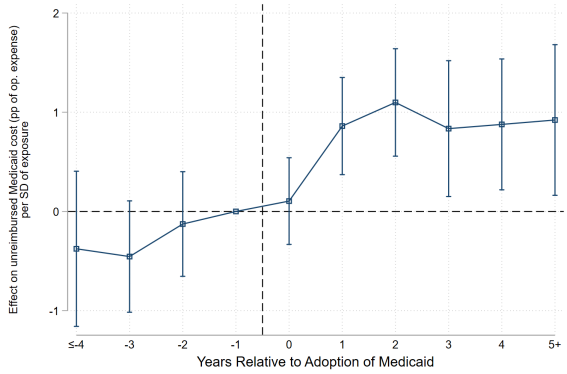
	(1)	(2)
Operating margin	-2.302**	-0.110
	(0.914)	(0.762)
Log inpatient discharges	0.006	0.004
	(0.020)	(0.021)
Occupancy rate	-0.647	-0.828
	(0.571)	(0.898)
Log net patient revenue	-0.022	0.012
	(0.017)	(0.027)
Log operating expense	0.001	0.020
	(0.013)	(0.029)
Exposure measure	Hospital burden	County eligibility
State × year FE	Y	Y
Exposure × year	Y	Y
County covariates	Y	Y

NOTES: Rural hospitals, 2011–2019, outcomes as in the main tables except that the payer-mix outcomes are shares of operating expense, in percent, as in the descriptive table. Each cell is the coefficient on the interaction of Post, defined as in the main tables, with a hospital's standardized exposure, so it is the change per standard deviation, from a separate regression of the row outcome. In column 1 exposure is the hospital's 2011–2013 uncompensated care share of operating expense, shrunk toward the mean of hospitals in the same state with the same critical access status; in column 2 it is the 2011–2013 share of the county's adults who were uninsured and below 138% of the federal poverty line. Every regression includes Post itself, hospital fixed effects, state-by-year fixed effects, interactions of exposure with each year, and the county covariates. Standard errors clustered by state in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

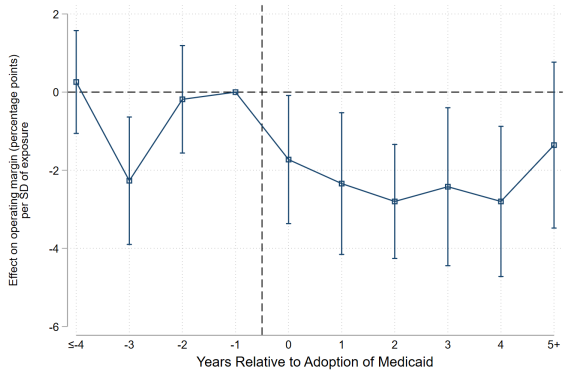
Figure B.6: Exposure Event Studies



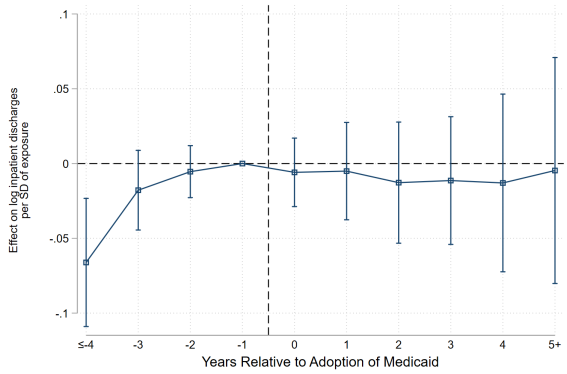
(a) Charity care for uninsured patients



(b) Unreimbursed Medicaid cost



(c) Operating margin



(d) Log inpatient discharges

NOTES: Each panel plots the coefficients on the interaction of a hospital's standardized 2011–2013 uncompensated care exposure with indicators for years relative to the hospital's first cost report that includes months after its state's expansion, omitting the year before. Every regression includes hospital fixed effects, state-by-year fixed effects, interactions of exposure with each calendar year, and the county covariates. Rural hospitals, 2011–2019. Vertical bars are 95% confidence intervals from standard errors clustered by state.

B.4 An Alternative Exposure Measure

Column 2 of Table B.4 and Table B.5 replaces the hospital's own pre-period uncompensated care burden with a county-level measure, the share of the county's adults who were uninsured and below 138 percent of the federal poverty line over 2011–2013, which measures the size of the newly eligible population. This measure produces no gradient on Medicaid shortfalls, margins, or volume, and only small ones on charity care and bad debt, which fall by an additional 0.27 and 0.40 percentage points per standard deviation (significant at the 10 and 5 percent levels). The two exposure measures are positively correlated (0.51), but the county measure describes the population around a hospital rather than the patients it actually treated, and the result indicates that the hospital's own pre-expansion burden, not the size of its county's newly eligible population, predicts which hospitals' charity care, Medicaid shortfalls, and margins changed. Three of the eight BEA regions contain

no state that had not expanded by 2019, so the region-by-year comparison in the paper rests on treatment timing within those regions and on non-expanding states in the Southeast, Plains, Great Lakes, Southwest, and Rocky Mountain regions.

B.5 Sample and Timing Checks

Rural status. Rural status in the Provider of Services file is recorded annually, so about 7 percent of hospitals move into or out of the rural sample at least once. If expansion caused hospitals to be reclassified, a sample defined year by year would be selected on treatment. Regressing an indicator for leaving the rural sample on expansion gives a coefficient of 0.0043 with a standard error of 0.0095, so it does not, but I fix each hospital's status at its 2011–2013 value regardless.

Facility linking. A hospital that converts to critical access status receives a new Medicare certification number, as do hospitals in some mergers; 237 facilities in the full cost report panel span more than one number, 104 of them through critical access conversions. Without linking, each conversion would appear as one hospital exiting and another entering with no pre-expansion history, which would remove exactly the hospitals whose 2011–2013 exposure the design depends on. I link predecessor and successor numbers and treat the facility as the unit of analysis.

Verification checks. Table B.6 reruns the preferred column 2 on several samples. Critical access hospitals, which Medicare pays on the basis of cost and which receive no Medicare uncompensated care payments, had no reason to change how they report charity care when CMS began using these reports to allocate those payments in fiscal year 2018. They show a 53% decline in charity care, against 67% at the other rural hospitals, and margins that rise by 1.9 and 2.4 points. A balanced panel of hospitals that filed in all nine years, and a sample without the three regions in which every state had expanded by 2019, give nearly identical estimates. Expansion did not change whether a hospital reported any charity care or any unreimbursed Medicaid cost. Hospitals in non-adopting states were more likely to stop filing before 2019 (8.7 against 3.7 percent, or 3.8 percentage points with region fixed effects), more than half of them through closure, which is why the balanced panel matters, and it changes little. Applying the payer-mix estimates from one stacked regression to 2013 means across all filers, with a hospital that reports none of an item counted as zero, gives net relief of $545,000 per hospital, with a 95% confidence interval of $320,000 to $769,000. Two hospitals in the U.S. Virgin Islands carry a CMS state code that the hospital roster maps to Texas, and I drop them.

Table B.6: Verification Checks, Rural Hospital Panel

	Log charity care, uninsured	Log non-Medicare bad debt	Log unreimb. Medicaid cost	Operating margin	Log inpatient discharges
Panel A: Column 2 on different samples					
(a) All rural hospitals (column 2)	-0.891***	-0.337***	0.180*	2.228***	-0.027
	(0.153)	(0.046)	(0.099)	(0.703)	(0.019)
Hospital-years	18,118	19,234	13,488	19,708	19,999
(b) Critical access hospitals	-0.750***	-0.313***	0.152	1.914**	-0.048*
	(0.147)	(0.055)	(0.103)	(0.924)	(0.027)
Hospital-years	10,679	11,435	8,329	11,732	11,834
(c) Other rural hospitals	-1.116***	-0.395***	0.237*	2.409**	0.011
	(0.169)	(0.048)	(0.124)	(1.177)	(0.022)
Hospital-years	7,439	7,799	5,159	7,976	8,165
(d) Filed in all nine years	-0.896***	-0.341***	0.166*	1.983***	-0.034*
	(0.155)	(0.047)	(0.099)	(0.694)	(0.019)
Hospital-years	17,458	18,473	13,020	18,888	19,042
(e) Excluding fully expanded regions	-0.908***	-0.341***	0.187*	2.227***	-0.031
	(0.165)	(0.049)	(0.107)	(0.752)	(0.020)
Hospital-years	15,479	16,466	11,319	16,877	17,135
Panel B: Other checks					
	Estimate	(S.E.)	Mean		
Reports any charity care (pp)	-1.346	(1.433)	90.59		
Reports any unreimbursed Medicaid cost (pp)	3.208	(2.887)	67.89		
Unreimbursed Medicaid cost, % of operating expense	0.744***	(0.198)	3.75		
Stopped filing before 2019 (pp), hospital level	-3.846***	(1.380)	8.73		
Net payer-mix relief per hospital (thousands of dollars)	545	(115)			

NOTES: Rural hospitals, 2011–2019, as in the main hospital tables. Each estimate in Panel A and the first three rows of Panel B is the coefficient on Post, the share of the hospital's cost-reporting period after its state's expansion date, from a regression with hospital fixed effects and BEA region-by-year fixed effects (column 2 of the main tables). For outcomes in logs, the percent effect of a coefficient b is $100(e^b - 1)$. Row (b) keeps critical access hospitals, which Medicare pays on the basis of cost and which receive no Medicare uncompensated care payments, so they had no reason to change how they report charity care when CMS began using these reports to allocate those payments in fiscal year 2018. Row (d) keeps hospitals that filed a cost report in every year from 2011 to 2019. Row (e) drops the Far West, the Mideast, and New England, where every state had expanded by 2019. In Panel B, the extensive-margin rows use every filed cost report with positive operating expense, and the means are percentages of those hospital-years. The pre-adoption event-study coefficients for the Medicaid shortfall share are jointly insignificant ($p = 0.96$). The attrition row regresses an indicator for a hospital's last cost report falling before 2019 on its state's expansion by 2019, with one observation per hospital and BEA region fixed effects; its mean is the rate in non-expansion states (3.7% in expansion states). Net relief applies the three payer-mix estimates from one stacked regression to 2013 means across all filers in expansion states, with a blank item counted as zero, in 2019 dollars, with a 95% confidence interval of 320 to 769 thousand dollars by the delta method. Standard errors clustered by state in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Window. The cost report data run past 2019, but I end the hospital panel there. The later years include the pandemic and the federal relief payments that accompanied it, which reached hospitals in expanding and non-expanding states on terms unrelated to Medicaid, and ending in 2019 also gives the hospital panel the same treatment window and control states as the state panel.

Closure. I do not estimate effects on hospital closure. There are 109 rural closures in the panel between 2011 and 2024, of which only 33 occur after expansion in an adopting state and only 15 before 2020. With so few events, any estimate would rest on a handful of hospitals.

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